

CATEGORICITY

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1. Uniqueness, completeness and categoricity

Description of an object in a language
– informally

Naive assumption on a language: *we can describe an object of our interest fully and completely, uniquely determining the object.*

Newtonian physics: such a description, a theory, is possible.

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What about **modern physics**?

In a formal logic we have

Completeness and *Categoricity*.

Completeness = 'fully and completely'
in terms of the formal language itself.

Categoricity = unique interpretation in
reality.

Categoricity – easier *acceptable* to
practical mathematicians and physicists.

Completeness – easier *achievable* on
the theoretical level.

Do the objects described by a formal theory exist in reality?

Depends on the type of the description, *complete* or *categorical*.

2. Categoricity. What categoricity?

Must not be based on a list of all possible configuration in all possible locations of the structure.

We want a *concise* description of a *large* structure.

Thus the (absolute) categoricity for first order languages is uninteresting.

Definition A structure $\mathbf{M}$ in a language L is said to have λ -**categorical** theory if there is exactly one, up to isomorphism, structure of cardinality λ satisfying the L -theory $\text{Th}(\mathbf{M})$ of $\mathbf{M}$.

Theorem (M.Morley, 1964 confirming
a conjecture by J.Los)

*If a countable first-order T is λ -categorical
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Theorem (S.Shelah, 1983)

If an $L_{\omega_1, \omega}$ -sentence is categorical in $\aleph_n$ for all n then it is categorical (and has models) in all infinite λ .

3. Stability, homogeneity and smoothness.

Theorem (M.Morley, S.Shelah)
Categoricity implies stability

Stability = dimension theory plus highly homogeneous models in all cardinalities).

Thesis *This is a weak form of smoothness*

4. Trichotomy Conjecture

Classical first-order λ -categorical structures for **uncountable** λ :

- (1) **Trivial** structures (= only)
- (2) **Linear structures** (Abelian divisible torsion-free groups; Vector spaces over a countable division ring, ...)
- (3) **Algebraically closed fields.**

One can construct more complicated structures over the basic ones preserving the property of categoricity, e.g.

Algebraic groups

$$\mathrm{GL}(n, \mathbb{C}), \mathrm{PGL}(n, \mathbb{C}), \dots$$

More generally,
complex **algebraic varieties** $V \subseteq \mathbb{C}^n$
equipped with polynomially defined relations

$$p(\bar{x}_1, \dots, \bar{x}_m) = 0$$

in $n \times m$ variables).

Observation – example: **Compact complex spaces** in **the natural** language are ω -stable of finite Morley rank.

If a compact complex space is Kähler it is saturated.

Trichotomy Conjecture: An uncountably categorical structure must be *classical*.

Trichotomy Conjecture (Z. 1982)

Given a f -o uncountably categorical structure $\mathbf{M}$ one and only one of the following holds:

- (i) the geometry of $\mathbf{M}$ is **trivial**;
- (ii) the geometry of $\mathbf{M}$ is isomorphic to an **affine or projective** geometry over a countable division ring;
- (iii) $\mathbf{M}$ is a structure of **algebraic geometry** over an algebraically closed field K .
($\mathbf{M}$ is definably equivalent to the field K .)

The Trichotomy conjecture

refuted in general (E.Hrushovski, 1989)

proved under extra **Zariski** assumptions
(E.Hrushovski and B.Zilber, 1994)

Also, proved in the **o-minimality** –
real algebraic geometry context
(Y.Peterzil, S.Starchenko, 1996).

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In the Zariski context

Pre-smoothness assumption on M
(**dimension theorem**):

For any **closed irreducible**

$$S_1, S_2 \subseteq M^n$$

$$\dim S_1 \cap S_2 \geq \dim S_1 + \dim S_2 - \dim M^n$$

component-wise.

Hrushovski's counter-examples

Given a class of structures $\mathbf{M}$ with a dimension notions ∂_1 , and ∂_2 we want to consider a *new function* f on $\mathbf{M}$.

On $(\mathbf{M}, f)$ introduce a **predimension**

$$\delta(X) = \partial_1(X \cup f(X)) - \partial_2(X).$$

Consider structures $(\mathbf{M}, f)$ which satisfy the **Hrushovski inequality**:

$$\delta(X) \geq 0 \text{ for any finite } X \subset \mathbf{M}.$$

Amalgamate all such structures to get a *universal and homogeneous* structure in the class.

The resulting structure $(\tilde{\mathbf{M}}, f)$ will have a good dimension notion and a nice geometry.

Observation:

If $\mathbf{M}$ is a field and we want $f = \text{ex}$ to be a group homomorphism

$$\text{ex}(x_1 + x_2) = \text{ex}(x_1) \cdot \text{ex}(x_2)$$

then the corresponding predimension must be

$$\delta(X) = \text{tr.deg}(X \cup \text{ex}(X)) - \text{lin.dim}_{\mathbb{Q}}(X) \geq 0.$$

The Hrushovski inequality, in the case of the complex numbers and $\text{ex} = \exp$, is equivalent to

$$\text{tr.deg}(x_1, \dots, x_n, e^{x_1}, \dots, e^{x_n}) \geq n$$

assuming that $x_1, \dots, x_n$ are linearly independent.

This is **the Schanuel conjecture**.

Pseudo-exponentiation

Consider the class of fields of characteristic 0 with a function ex :

$K_{\text{ex}} = (K, +, \cdot, \text{ex})$ satisfying

$$\text{EXP1: } \text{ex}(x_1 + x_2) = \text{ex}(x_1) \cdot \text{ex}(x_2)$$

$$\text{EXP2: } \ker \text{ex} = \pi\mathbb{Z}$$

Consider the subclass satisfying the Schanuel condition

$$\text{SCH : } \text{tr.deg}(X \cup \text{ex}(X)) - \text{lin.dim}_{\mathbb{Q}}(X) \geq 0.$$

Amalgamation process produces K_{ex} ,
an *Existentially Closed*

field with pseudo-exponentiation,

Existential Closedness property

EC: Any well-overdetermined system of equations in $+$, $\cdot$, ex has a solution in K_{ex} .

And

Countable Closure property

CC: 'Analytic' subsets of K^n of dimension 0 are countable.

Theorem *Given $\lambda > \aleph_0$, there is a unique model of axioms $\text{ACF}_0 + \text{EXP} + \text{SCH} + \text{EC} + \text{CC}$ of cardinality λ .*

This is a consequence of

Theorem A *The $L_{\omega_1, \omega}(Q)$ -sentence $\text{ACF}_0 + \text{EXP} + \text{SCH} + \text{EC} + \text{CC}$ is axiomatising a **quasi-minimal excellent class**.*

and

Theorem B (Essentially S.Shelah 1983)
A quasi-minimal excellent class has models and is categorical in any uncountable cardinality.

There are a series of further pseudo-analytic and analytic examples.

Theorem (A.Wilkie, P.Koiran, B.Z.)

There is an entire analytic function f which satisfies:

(i) Schanuel-type property (Hrushovski inequality)

$\text{tr.deg}(x_1, \dots, x_n, f(x_1), \dots, f(x_n)) \geq n$
for distinct $x_1, \dots, x_n \in \mathbb{C}$

(ii) existential closedness property,

(iii) the first-order theory T_f of $(\mathbb{C}, +, \cdot, f)$ is ω -stable,

(iv) the $L_{\omega_1, \omega}(Q)$ -sentence $T_f + \text{CC}$ is categorical in all uncountable cardinals.

Conclusions.

1. Taking into account λ -categoricity and stability for stronger languages we can extend the list of basic geometries:

Basic geometries of stability theory:

(1) **Trivial geometry**

(2) **Linear geometry**

(3) **Algebraic geometry**

(*) **“Classical analytic” geometries**
– *fusions* of (1)–(3).

2. We can predict for classical analytic geometries **Schanuel-type property**
– a Hrushovski inequality.

3. We can predict for classical analytic geometries the **existential closeness** property –
– Any non-overdetermined and free (from obvious contradictions) system of equations has a solution.

Universal covers of semi-abelian varieties

$$0 \longrightarrow \Lambda \xrightarrow{i} \mathbb{C}^g \xrightarrow{\exp} \mathbb{A}(\mathbb{C}) \longrightarrow 1,$$

where $\exp$ is an analytic homomorphism from the additive group $(\mathbb{C}^g, +)$ and $\Lambda = \mathbb{Z}^N$ is a discrete subgroup of $\mathbb{C}^g$, $\exp$ a group homomorphism.

Question Is the universal cover uniquely determined by algebraic data **only**?

Universal covers of semi-abelian varieties

$$0 \longrightarrow \Lambda \xrightarrow{i} V \xrightarrow{\text{ex}} \mathbb{A}(F) \longrightarrow 1$$

$(V, +)$ a group, F an algebraically closed field, $\mathbb{A}(F)$ the F -points of semiabelian variety with the structure induced by F , ex a group homomorphism.

Reformulation Is the obvious L_{ω_1, ω^-} -sentence Σ_A describing the sequence uncountably **categorical**?

Theorem C Σ_A is uncountably categorical iff the following arithmetic conditions hold for **good** fields k :

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(i) (**Galois action on torsion points**)
for all but finitely many prime p the group $\text{Gal}(\tilde{k} : k)$ acts on the Tate module $T_p(\mathbb{A})$ as $\text{GL}_N(\mathbb{Z}_p)$, and for remaining finite number of p the group acts as a subgroup of $\text{GL}_N(\mathbb{Z}_p)$ of finite index;

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(ii) (**Kummer theory and heights**)

given a mult-independent $a_1, \dots, a_n$ there is an $l \in \mathbb{N}$ such that for any $m \in \mathbb{N}$ s.t. $\mathbb{A}(k)$ contains ml -torsion

$$\begin{aligned} \text{Gal}(k(a_1^{\frac{1}{ml}}, \dots, a_n^{\frac{1}{ml}}) : k(a_1^{\frac{1}{l}}, \dots, a_n^{\frac{1}{l}})) \\ \cong (\mathbb{Z}/m\mathbb{Z})^{Nn}. \end{aligned}$$

Theorem C is a consequence of Keisler – Shelah theory of $L_{\omega_1, \omega}$ -categoricity (*excellency*).

Galois action on roots of unity, Kummer and height theories are known for some $\mathbb{A}$.

This implies

Theorem A *The $L_{\omega_1, \omega}(Q)$ -sentence $\text{ACF}_0 + \text{EXP} + \text{SCH} + \text{EC} + \text{CC}$ is axiomatising a **quasi-minimal excellent class**.*

Conclusion 4 Categoricity can predict strong **arithmetic** properties.

Analytic context

Can the universal cover of the multiplicative group (torus) F^* be *compactified* as an **analytic Zariski** structure

$$0 \longrightarrow \Lambda \xrightarrow{i} V \xrightarrow{\text{ex}} F^* \longrightarrow 1?$$

This leads to the theory of **toric geometry**.

Toric geometry is the main model for **string theory and mirror symmetry**.

Hope

Uniqueness - categoricity criterion
can help to find a true mathematical
model of physics.